

SSC CGL PREPARATION

Day 2 — Complete Study Notes

Quantitative Aptitude • Reasoning • English •
General Awareness

Fractions,
Decimals,
Surds &
Indices

Coding-
Decoding
(Mixed &
Matrix)

Vocabulary:
Synonyms &
Antonyms

Medieval
India
(Sultanate &
Mughals)

Table of Contents

1. Quantitative Aptitude — Fractions, Decimals, Surds & Indices
2. Reasoning — Coding-Decoding (Mixed & Matrix Coding)
3. English Language — Synonyms & Antonyms
4. General Awareness — Medieval India (Delhi Sultanate & Mughals)

Fractions, Decimals, Surds & Indices

Core rules, shortcuts, worked examples, and previous-year SSC CGL questions.

Part A — Fractions & Decimals

1. Fractions — Types

Type	Definition	Example
Proper Fraction	Numerator < Denominator	$3/7$, $5/9$, $11/13$
Improper Fraction	Numerator > Denominator	$7/3$, $11/5$, $17/4$
Mixed Fraction	Whole Number + Proper Fraction	$2\frac{1}{3}$, $5\frac{5}{8}$, $3\frac{3}{4}$
Like Fractions	Same Denominator	$2/7$, $5/7$, $9/7$
Unlike Fractions	Different Denominators	$2/5$, $3/7$, $4/9$
Equivalent Fractions	Same Value	$1/2 = 2/4 = 3/6 = 4/8$
Unit Fraction	Numerator = 1	$1/2$, $1/3$, $1/7$

2. Operations on Fractions

(i) Addition & Subtraction

Like fractions: $2/7 + 3/7 = 5/7$ | $5/9 - 2/9 = 3/9 = 1/3$

Unlike fractions: Find LCM of denominators → convert to equivalent fractions → add/subtract numerators.

Example: $2/3 + 3/4 \rightarrow \text{LCM}(3,4)=12 \rightarrow 8/12 + 9/12 = 17/12 = 1\frac{5}{6}$

(ii) Multiplication

$$a/b \times c/d = (a \times c)/(b \times d)$$

$$2/3 \times 4/5 = 8/15 \quad | \quad 3/4 \times 8/9 = 24/36 = 2/3$$

Cross-cancellation trick: $4/9 \times 3/8 \rightarrow$ cancel 4 & 8 $\rightarrow 1/9 \times 3/2 = 1/6$

(iii) Division

$$a/b \div c/d = a/b \times d/c = (a \times d)/(b \times c)$$

$$2/3 \div 4/5 = 2/3 \times 5/4 = 5/6 \quad | \quad 5/8 \div 15/16 = 5/8 \times 16/15 = 2/3$$

3. Important Fraction-Percentage-Decimal Shortcuts

Fraction	%	Decimal	Fraction	%	Decimal
1/2	50%	0.5	1/8	12.5%	0.125
1/3	33.33%	0.333...	3/8	37.5%	0.375
2/3	66.67%	0.666...	5/8	62.5%	0.625
1/4	25%	0.25	7/8	87.5%	0.875
3/4	75%	0.75	1/10	10%	0.1
1/5	20%	0.2	1/20	5%	0.05
2/5	40%	0.4	1/25	4%	0.04
3/5	60%	0.6	1/50	2%	0.02
4/5	80%	0.8	1/100	1%	0.01

Memorize: These conversions appear frequently in SSC CGL simplification questions.

4. Decimals — Types

Type	Definition	Example
Terminating Decimal	Ends after finite digits	0.25, 0.125, 2.75
Recurring (Non-terminating)	Repeats indefinitely	0.333..., 0.121212...
Pure Recurring	All digits repeat	$0.\overline{3}$, $0.\overline{12}$
Mixed Recurring	Some digits don't repeat	$0.1\overline{2}$, $0.234\overline{5}$

Conversion: Terminating Decimal → Fraction

Write the number without the decimal point over 10^n (n = number of decimal places).

$$0.25 = 25/100 = 1/4 \quad | \quad 0.125 = 125/1000 = 1/8 \quad | \quad 2.75 = 275/100 = 11/4 \quad | \quad 0.0002 = 2/10000 = 1/5000$$

Conversion: Pure Recurring Decimal → Fraction

$0.\overline{abc} = abc / 999$ (as many 9's as repeating digits)

$$0.\overline{3} = 3/9 = 1/3 \quad | \quad 0.\overline{12} = 12/99 = 4/33 \quad | \quad 0.\overline{123} = 123/999 = 41/333$$

Conversion: Mixed Recurring Decimal → Fraction

Numerator = (whole number incl. non-repeating & repeating digits) – (non-repeating part only)
 Denominator = as many 9's as repeating digits, followed by as many 0's as non-repeating digits

$$0.\overline{12} = (12-1)/90 = 11/90$$

$$0.\overline{345} = (345-3)/990 = 342/990 = 19/55$$

$$0.\overline{123} = (123-1)/990 = 122/990 = 61/495$$

Tip: Always simplify the fraction to lowest terms.

5. Operations on Decimals

Operation	Rule	Example
Addition/ Subtraction	Align decimal points, pad with zeros	$2.35 + 3.70 = 6.05$ $5.20 - 3.45 = 1.75$
Multiplication	Multiply as whole numbers, count total decimal places	$2.5 \times 3.4 = 8.5$ $0.02 \times 0.3 = 0.006$
Division	Shift decimal so denominator is a whole number	$2.5 \div 0.5 = 5$ $0.048 \div 0.02 = 2.4$

Part B — Surds

1. Definition

A **surd** is an irrational number that cannot be simplified to remove the radical sign. Example: $\sqrt{2}$, $\sqrt{3}$, $\sqrt[3]{5}$ are surds; $\sqrt{4} = 2$ and $\sqrt[3]{8} = 2$ are **not** surds.

2. Laws of Surds

Law	Formula	Example
Product Law	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{2} \times \sqrt{3} = \sqrt{6}$
Quotient Law	$\sqrt{a} \div \sqrt{b} = \sqrt{a/b}$	$\sqrt{8} \div \sqrt{2} = \sqrt{4} = 2$
Power Law	$(\sqrt{a})^n = a^{(n/2)}$	$(\sqrt{5})^2 = 5$
Equality Law	$\sqrt{x} = \sqrt{y} \Rightarrow x = y$	$\sqrt{x} = \sqrt{7} \Rightarrow x = 7$
Mixed Surd	$a\sqrt{b}$	$3\sqrt{5}$, $2\sqrt{7}$
Pure Surd	$\sqrt{a}$	$\sqrt{11}$, $\sqrt{13}$

3. Operations on Surds

Addition/Subtraction (like surds only):

$$3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2} \quad | \quad 7\sqrt{5} - 2\sqrt{5} = 5\sqrt{5} \quad | \quad 3\sqrt{2} + 4\sqrt{3} - \sqrt{2} + 2\sqrt{3} = 2\sqrt{2} + 6\sqrt{3}$$

Note: $3\sqrt{2} + 4\sqrt{3}$ cannot be simplified further (unlike surds).

Multiplication: $3\sqrt{2} \times 5\sqrt{3} = 15\sqrt{6}$ | $(2\sqrt{3})(4\sqrt{3}) = 24$ | $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = 3 - 2 = 1$

Division: $6\sqrt{15} \div 3\sqrt{3} = 2\sqrt{5}$ | $\sqrt{12} \div \sqrt{3} = \sqrt{4} = 2$

4. Rationalization of Surds

Rationalization removes the surd from the denominator.

Type 1 — Single surd denominator: multiply by $\sqrt{a}/\sqrt{a}$.

$$1/\sqrt{3} \times \sqrt{3}/\sqrt{3} = \sqrt{3}/3 \quad | \quad 5/\sqrt{2} \times \sqrt{2}/\sqrt{2} = 5\sqrt{2}/2$$

Type 2 — Binomial surd denominator ($a+\sqrt{b}$): multiply by conjugate ($a-\sqrt{b}$).

$$1/(2+\sqrt{3}) \rightarrow (2-\sqrt{3})/(4-3) = 2-\sqrt{3}$$

$$3/(\sqrt{5}-\sqrt{2}) \rightarrow 3(\sqrt{5}+\sqrt{2})/(5-2) = \sqrt{5}+\sqrt{2}$$

Type 3 — Denominator ($a+b\sqrt{c}$): multiply by conjugate ($a-b\sqrt{c}$).

$$1/(3+2\sqrt{2}) \rightarrow (3-2\sqrt{2})/(9-8) = 3-2\sqrt{2}$$

5. Comparison of Surds

Compare $\sqrt{3}$ and $\sqrt[3]{5}$: convert to same root $\rightarrow 3^{(3/6)}=27^{(1/6)}$, $5^{(2/6)}=25^{(1/6)}$. Since $27 > 25$, $\sqrt{3} > \sqrt[3]{5}$.

Compare $2\sqrt{3}$ and $3\sqrt{2}$: $2\sqrt{3} = \sqrt{12}$, $3\sqrt{2} = \sqrt{18}$. Since $18 > 12$, $3\sqrt{2} > 2\sqrt{3}$.

Part C — Indices

1. Laws of Indices

Law	Formula	Example
Product Rule	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7$
Quotient Rule	$a^m \div a^n = a^{m-n}$	$5^6 \div 5^2 = 5^4$
Power Rule	$(a^m)^n = a^{m \times n}$	$(3^2)^3 = 3^6$
Power of Product	$(ab)^m = a^m \times b^m$	$(2 \times 3)^2 = 2^2 \times 3^2$
Power of Quotient	$(a/b)^m = a^m/b^m$	$(2/3)^2 = 2^2/3^2$
Zero Exponent	$a^0 = 1$	$7^0=1, 100^0=1$
Negative Exponent	$a^{-m} = 1/a^m$	$2^{-3} = 1/8$
Fractional Exponent	$a^{(m/n)} = \sqrt[n]{a^m}$	$8^{(2/3)} = \sqrt[3]{(8^2)} = 4$
Base Change	$a^m = b^n \Rightarrow a = b^{(n/m)}$	$2^x = 8 \Rightarrow x = 3$

2. Important Algebraic Identities

Identity	Formula
Difference of Squares	$a^2 - b^2 = (a+b)(a-b)$
Sum of Cubes	$a^3 + b^3 = (a+b)(a^2-ab+b^2)$
Difference of Cubes	$a^3 - b^3 = (a-b)(a^2+ab+b^2)$

Perfect Square	$(a+b)^2 = a^2+2ab+b^2$
Perfect Square	$(a-b)^2 = a^2-2ab+b^2$
Cube Expansion	$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$
Cube Expansion	$(a-b)^3 = a^3-3a^2b+3ab^2-b^3$

Part D — Previous Year Questions (Surds & Indices)

Q1 Find the value of $(256)^{0.16} \times (256)^{0.09}$.

(a) 4 (b) 16 (c) 64 (d) 256.25

$(256)^{0.16+0.09} = 256^{0.25} = (2^8)^{(1/4)} = 2^2 = 4$. **Answer: (a) 4**

Q2 If $2^x = 8^y = 64$, find x/y .

(a) 1 (b) 2 (c) 3 (d) 1.5

$2^x=64=2^6 \Rightarrow x=6$. $8^y=64 \Rightarrow 2^{3y}=2^6 \Rightarrow y=2$. $x/y=6/2=3$. **Answer: (c) 3**

Q3 Simplify: $[(216)^{(2/3)} \times (25)^{(1/2)}] \div (16)^{(3/4)}$

(a) 10 (b) 15 (c) 20 (d) 22.5

$(216)^{(2/3)}=6^2=36$; $(25)^{(1/2)}=5$; $(16)^{(3/4)}=2^3=8$. Result = $(36 \times 5)/8 = 22.5$. **Answer: (d) 22.5**

Q4 Find the value of $\sqrt{20} + \sqrt{45} - \sqrt{80}$.

(a) $2\sqrt{5}$ (b) $3\sqrt{5}$ (c) $\sqrt{5}$ (d) $5\sqrt{5}$

$\sqrt{20}=2\sqrt{5}$, $\sqrt{45}=3\sqrt{5}$, $\sqrt{80}=4\sqrt{5}$. Sum = $(2+3-4)\sqrt{5} = \sqrt{5}$. **Answer: (c) $\sqrt{5}$**

Q5 If $5^x = 125$ and $2^y = 32$, find $x+y$.

(a) 5 (b) 7 (c) 8 (d) 10

$5^x=125=5^3 \Rightarrow x=3$. $2^y=32=2^5 \Rightarrow y=5$. $x+y=8$. **Answer: (c) 8**

Q6 Rationalize the denominator: $1/(3+\sqrt{5})$

(a) $(3-\sqrt{5})/4$ (b) $(3+\sqrt{5})/4$ (c) $(3-\sqrt{5})/2$ (d) $(3+\sqrt{5})/2$

$1/(3+\sqrt{5}) \times (3-\sqrt{5})/(3-\sqrt{5}) = (3-\sqrt{5})/(9-5) = (3-\sqrt{5})/4$. **Answer: (a) $(3-\sqrt{5})/4$**

Q7 Simplify: $(2^{n+4} - 2 \times 2^n) \div (2 \times 2^{n+3})$

(a) $1/2$ (b) 1 (c) $7/8$ (d) $3/4$

$= [2^{n+4} - 2^{n+1}] / 2^{n+4} = 1 - 2^{-3} = 1 - 1/8 = 7/8$. **Answer: (c) $7/8$**

Q8 If $\sqrt{3} = 1.732$, find the value of $(\sqrt{3}-1)/(\sqrt{3}+1)$.

(a) 0.268 (b) 0.732 (c) 0.366 (d) 0.134

Rationalize: $(\sqrt{3}-1)^2/2 = (4-2\sqrt{3})/2 = 2-\sqrt{3} = 2-1.732 = 0.268$. **Answer: (a) 0.268**

Q9 Find the value of $125^{(2/3)} \times 25^{(1/2)}$.

(a) 125 (b) 625 (c) 25 (d) 3125

$125^{(2/3)} = (5^3)^{(2/3)} = 5^2 = 25$. $25^{(1/2)} = 5$. $25 \times 5 = 125$. **Answer: (a) 125**

Q10 If $a = 2 + \sqrt{3}$, find $a + 1/a$.

(a) 4 (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) 2

$1/a = 1/(2+\sqrt{3}) = 2-\sqrt{3}$. $a + 1/a = (2+\sqrt{3}) + (2-\sqrt{3}) = 4$. **Answer: (a) 4**

Q11 If $2^x = 8^y$, find $x:y$.

(a) 1:3 (b) 3:1 (c) 2:3 (d) 3:2

$2^x = 8^y = 2^{3y} \Rightarrow x = 3y \Rightarrow x:y = 3:1$. **Answer: (b) 3:1**

Q12 Simplify: $\sqrt{5} + 2\sqrt{5} - 3\sqrt{5}$

(a) 0 (b) $\sqrt{5}$ (c) $2\sqrt{5}$ (d) $3\sqrt{5}$

$(1+2-3)\sqrt{5} = 0$. **Answer: (a) 0**

Additional Practice Questions

#	Question	Answer
Q13	Find $\sqrt{0.09} + \sqrt{0.16}$	0.7 (0.3 + 0.4)
Q14	Simplify $(81)^{(3/4)}$	27
Q15	If $3^x = 81$ and $4^y = 64$, find $x+y$	7
Q16	Rationalize $2/(1+\sqrt{3})$	$\sqrt{3} - 1$
Q17	Find $\sqrt{48} - \sqrt{27} + \sqrt{12}$	$3\sqrt{3}$

Key Takeaways for the Exam

- Memorize fraction-decimal-percentage conversions ($1/2 = 0.5 = 50\%$, etc.)

- **BODMAS** is crucial for simplification questions.
- **Rationalization** appears frequently — practice it well.
- Laws of indices problems are direct and scoring.
- Convert recurring decimals to fractions quickly using the 9's-and-0's rule.
- Only **like surds** can be added or subtracted.
- Always simplify fractions to their lowest terms.

Coding-Decoding: Mixed & Matrix Coding

Pattern recognition techniques, letter-shift rules, matrix conventions, and PYQs.

Coding-Decoding is one of the most scoring topics in the SSC CGL Reasoning section — typically **3-5 questions** in Tier 1. The key is to identify the coding pattern quickly and apply it consistently.

Part A — Mixed Coding

In Mixed Coding, the code combines multiple rules (shifting, reversal, vowel/consonant treatment) rather than one simple pattern.

Type	Description	Example
Letter Shifting	Moving letters forward/backward by fixed positions	A→C (+2)
Reverse Alphabet	Using opposite letters (A↔Z, B↔Y)	A→Z, B→Y
Vowel-Consonant Shift	Vowels shift differently than consonants	A→D (+3), B→D (+2)
Position Reversal	Reversing order of letters	GOOD→DOOG
Mathematical Operation	Adding/subtracting alphabet positions	A(1)→B(2), +1

1. Simple Letter Shifting

Forward shift (+n): If CAT→FDW (each letter +3), then DOG → D+3=G, O+3=R, G+3=J → **GRJ**

Backward shift (−n): If DOG→ALD (each letter −3), then CAT → C−3=Z, A−3=X, T−3=Q → **ZXQ**

Mixed/uniform shift: If APPLE→CRRNG (+2 each), then MANGO → **OCPIQ**

2. Reverse Alphabet Coding

Key rule: Opposite letters sum to 27 (A↔Z, B↔Y, C↔X, D↔W, E↔V, F↔U, G↔T, H↔S, I↔R, J↔Q, K↔P, L↔O, M↔N). Formula: reverse position = 27 − original position.

If GOOD→TLLW (opposite letters), then BAD → B↔Y, A↔Z, D↔W → **YZW**

3. Letter + Number Coding

A=1, B=2 ... Z=26. GOOD → G=7, O=15, O=15, D=4 → **7-15-15-4**
Reverse: 20-8-9-19 → T,H,I,S → **THIS**

4. Mixed Coding with Vowel Shift

Vowels and consonants shift by different amounts. Example rule: vowels +2, consonants +1.
APPLE → A(+2)=C, P(+1)=Q, P(+1)=Q, L(+1)=M, E(+2)=G → **CQQMG**
Applying the same rule to ORANGE → **QSCOHG**

5. General Approach for Irregular / Mixed-Shift Patterns

Some SSC CGL questions use a shift sequence that is *not constant* across letters (e.g. +1, -1, +1, +1, -1, -1). To solve these:

1. Write the alphabet position of each letter in the coded word and the original word.
2. Find the difference (shift) for each letter position, in order.
3. Apply the **same sequence of shifts** to the new word, letter by letter, wrapping around the alphabet (mod 26) if needed.

Example If MONKEY is coded as NPOLEZ, how will TIGER be coded?

M→N (+1), O→P (+1), N→O (+1), K→L (+1), E→E (0), Y→Z (+1)
Pattern: +1, +1, +1, +1, 0, +1
TIGER (5 letters) → T(+1)=U, I(+1)=J, G(+1)=H, E(+1)=F, R(+0)=R
Answer: UJHFR

Example If 'A' is coded as 'Z', 'B' as 'Y', 'C' as 'X' (opposite-letter rule), how will GOOD be coded?

G↔T, O↔L, O↔L, D↔W → **Answer: TLLW**

Example If JUNE is coded as HSLC, how will AUGUST be coded?

J→H(-2), U→S(-2), N→L(-2), E→C(-2) → pattern is -2 for every letter.
AUGUST → A(-2)=Y, U(-2)=S, G(-2)=E, U(-2)=S, S(-2)=Q, T(-2)=R
Answer: YSESQR

Example If HELLO is coded as JGNNQ, how will WORLD be coded?

H→J(+2), E→G(+2), L→N(+2), L→N(+2), O→Q(+2) → pattern is +2 for every letter.
WORLD → W(+2)=Y, O(+2)=Q, R(+2)=T, L(+2)=N, D(+2)=F
Answer: YQTNF

Exam tip: Always verify your shift pattern against every letter of the sample word before applying it — a single miscounted letter throws off the whole answer. If the shift sequence looks irregular, double-check whether it is actually constant (a common source of error).

Part B — Matrix Coding

In Matrix Coding, a table of letters is given with row and column labels (usually 0–4, or two matrices numbered 0–4 and 5–9). Each letter is represented by a (row, column) coordinate pair, and you must identify which option correctly encodes the target word.

Standard Single Matrix (0–4)

Matrix	0	1	2	3	4
0	A	B	C	D	E
1	F	G	H	I	J
2	K	L	M	N	O
3	P	Q	R	S	T
4	U	V	W	X	Y

Reading convention: the first digit = row, second digit = column. E.g. 02 = Row 0, Col 2 = C. So **CAT** → C=02, A=00, T=34 → **02 00 34**.

Important: SSC CGL sometimes reverses the convention (column-then-row). Always confirm from the example given in the question stem before solving — the same pair (e.g. "20") can represent two different letters under the two conventions.

Two-Matrix Format (Most Common in SSC CGL)

Matrix I uses rows/columns 0–4; Matrix II uses rows/columns 5–9. Every letter can be represented by a coordinate from *either* matrix, giving two valid codes per letter.

Matrix I	0	1	2	3	4
0	A	B	C	D	E
1	F	G	H	I	J
2	K	L	M	N	O
3	P	Q	R	S	T
4	U	V	W	X	Y

Matrix II	5	6	7	8	9
5	Z	A	B	C	D
6	E	F	G	H	I
7	J	K	L	M	N
8	O	P	Q	R	S
9	T	U	V	W	X

Worked Example Which set of numbers represents the word "CARE"?

(a) 02, 00, 13, 04 (b) 02, 00, 32, 04 (c) 02, 00, 31, 04 (d) 02, 00, 32, 05

C = Row 0, Col 2 = 02 (Matrix I) or Row 5, Col 8 = 58 (Matrix II)

A = Row 0, Col 0 = 00 (Matrix I) or Row 5, Col 6 = 56 (Matrix II)

R = Row 3, Col 2 = 32 (Matrix I) or Row 8, Col 8 = 88 (Matrix II)

E = Row 0, Col 4 = 04 (Matrix I) or Row 6, Col 5 = 65 (Matrix II)

So CARE = 02, 00, 32, 04 — matches option (b) exactly.

Answer: (b) 02, 00, 32, 04

How to Solve Matrix Coding Quickly

1. Write out every letter of the target word.
2. Find all possible (row, column) pairs for each letter — one from Matrix I, one from Matrix II.
3. Scan the options: check the **first** pair only. Eliminate any option whose first pair doesn't match a possible code for the first letter.
4. Repeat for the second, third, and fourth letters until only one option survives.

Part C — Exam Tips for Coding-Decoding

1. Identify the Pattern Quickly

Pattern Type	How to Identify
Simple Shifting	All letters move by the same number of positions
Mixed Shifting	Different shift amounts at different letter positions
Reverse Alphabet	Letters are replaced by their opposites (sum = 27)
Letter + Number	Letters are coded as their alphabet position numbers
Matrix Coding	A row-column matrix/grid is given in the question

2. Use "EJOTY" for Quick Letter Positions

Letter	Position
E	5
J	10
O	15
T	20
Y	25

Memory trick: "EJOTY" sounds like "Egypt" with a 'y' at the end — use it as anchor points to quickly count any letter's position.

3. Opposite Letters (Sum = 27)

Letter	Opposite	Letter	Opposite
A	Z	N	M
B	Y	O	L
C	X	P	K
D	W	Q	J
E	V	R	I
F	U	S	H
G	T	T	G
H	S	U	F
I	R	V	E
J	Q	W	D
K	P	X	C
L	O	Y	B
M	N	Z	A

4. Matrix Coding Tips

- Always check the row-column vs column-row convention stated in the question.
- Write the coordinate pairs for every letter of the target word first.
- Check options one at a time — eliminate quickly using the first letter's pair.
- SSC CGL typically uses **two matrices** (0-4 and 5-9), giving each letter two valid codes.

Quick Practice Questions

#	Question	Answer
Q1	If 'ORANGE' is coded as 'PSBOHF', find the code for 'APPLE'.	BQQMF
Q2	If A=1, B=2 ... Z=26, code the word 'INDIA'.	9-14-4-9-1
Q3	Using the standard 0-4 matrix above, what is the code for 'BAT'?	01, 00, 34 (or 57, 56, 95 in Matrix II)
Q4	If 'GOOD' is coded as 'HPPE' (each letter +1), find the code for 'BAD'.	CBE
Q5	If 'MORNING' is coded as 'LQMOHMF', find the code for 'EVENING'.	DUDMFMF

Vocabulary: Synonyms & Antonyms

High-frequency SSC CGL word lists (A-Z), previous-year questions, and exam strategy.

Synonyms and Antonyms are a crucial, high-scoring part of the SSC CGL English section — typically **3-5 questions** across Tier 1 and Tier 2.

Part A — Synonyms (A-Z Reference)

Types of Synonyms

Type	Description	Example
Exact Synonyms	Identical meaning	Begin = Commence
Near Synonyms	Similar but not identical	Laugh = Giggle, Chuckle
Contextual Synonyms	Same meaning in a specific context	Run = Operate (machine)

Most Important Synonyms (A-Z)

Word — Synonym(s) — Meaning. Focus on bolded high-frequency exam words.

Word	Synonym(s)	Meaning
Abate	Subside, Decrease, Diminish	To become less intense
Abdicate	Renounce, Resign, Relinquish	To give up power/position
Aberration	Deviation, Divergence, Anomaly	Deviation from normal
Abhor	Detest, Loathe, Despise	To regard with disgust
Abstain	Refrain, Desist, Withhold	To refrain from doing
Abstruse	Obscure, Profound, Recondite	Difficult to understand
Abundant	Plentiful, Ample, Copious	Existing in large quantities
Acclaim	Praise, Applaud, Commend	To praise enthusiastically
Acute	Sharp, Keen, Intense	Very serious or severe
Adept	Proficient, Expert, Skilled	Very skilled at something
Admonish	Reprimand, Rebuke, Warn	To warn or reprimand
Adversary	Opponent, Rival, Enemy	One who opposes
Affable	Friendly, Amiable, Cordial	Pleasant and easy to talk to
Affluent	Wealthy, Rich, Prosperous	Having great wealth
Aggravate	Worsen, Exacerbate, Intensify	To make worse
Alacrity	Eagerness, Enthusiasm, Readiness	Brisk and cheerful readiness

Allege	Claim, Assert, Maintain	To state without proof
Ambiguous	Vague, Obscure, Equivocal	Open to multiple interpretations
Ameliorate	Improve, Enhance, Upgrade	To make better
Amicable	Cordial, Harmonious, Peaceable	Characterized by friendliness
Anomaly	Aberration, Irregularity, Deviation	Something that deviates from normal
Apathy	Indifference, Unconcern, Lethargy	Lack of interest
Appease	Pacify, Placate, Calm	To satisfy or relieve
Arbitrary	Capricious, Random, Despotic	Based on whim rather than reason
Ardent	Passionate, Fervent, Zealous	Very enthusiastic
Arduous	Difficult, Strenuous, Laborious	Requiring great effort
Astute	Shrewd, Clever, Canny	Having sharp insight
Audacious	Bold, Daring, Intrepid	Showing willingness to take risks
Auspicious	Favorable, Propitious, Lucky	Conducive to success
Avarice	Greed, Cupidity, Covetousness	Extreme greed for wealth

Word	Synonym(s)	Meaning
Baffle	Confuse, Bewilder, Perplex	To confuse completely
Barbaric	Brutal, Savage, Primitive	Cruel and uncivilized
Benevolent	Kind, Generous, Charitable	Well-meaning and kind
Benign	Gentle, Harmless, Kind	Not harmful in effect
Bias	Prejudice, Preconception, Partiality	Inclination or prejudice
Bizarre	Strange, Weird, Eccentric	Very unusual
Blunt	Frank, Direct, Abrupt	Without sharpness or subtlety
Boon	Blessing, Advantage, Benefit	A timely benefit
Brittle	Breakable, Fragile, Crisp	Hard but easily broken

Word	Synonym(s)	Meaning
Calamity	Disaster, Catastrophe, Tragedy	An event causing great damage
Callous	Insensitive, Heartless, Cruel	Showing no concern
Candid	Frank, Honest, Straightforward	Truthful and direct
Capricious	Fickle, Whimsical, Erratic	Given to sudden change
Chastise	Punish, Reprimand, Scold	To criticize or punish
Chronic	Persistent, Constant, Unending	Long-lasting
Coerce	Compel, Force, Constrain	To force by threat

Complacent	Self-satisfied, Smug, Content	Smugly satisfied
Concede	Admit, Acknowledge, Grant	To yield or give in
Condone	Forgive, Overlook, Excuse	To overlook an offense
Contempt	Disdain, Scorn, Disrespect	The feeling of being scornful
Contrite	Repentant, Sorry, Remorseful	Feeling remorse
Copious	Plentiful, Abundant, Profuse	In large quantities
Cunning	Sly, Shrewd, Devious	Clever in deceiving

Word	Synonym(s)	Meaning
Debilitate	Weaken, Enervate, Sap	To make weak
Deplete	Exhaust, Drain, Consume	To use up
Deplore	Regret, Lament, Condemn	To feel or express sorrow
Desolate	Barren, Abandoned, Forlorn	Deprived of inhabitants
Despise	Detest, Loathe, Hate	To regard with contempt
Deter	Discourage, Prevent, Hinder	To discourage from acting
Devastate	Destroy, Ruin, Ravage	To cause great damage
Diligent	Industrious, Assiduous, Persistent	Hard-working
Disparage	Belittle, Deride, Demean	To speak disrespectfully about

Word	Synonym(s)	Meaning
Eager	Keen, Enthusiastic, Anxious	Wanting very much
Elaborate	Detailed, Intricate, Complex	Detailed and complex
Eloquent	Expressive, Articulate, Fluent	Fluent and persuasive
Eminent	Distinguished, Prominent, Illustrious	Outstanding in reputation
Emphatic	Forceful, Assertive, Vehement	Expressed with force
Eradicate	Eliminate, Erase, Destroy	To destroy completely
Exuberant	Enthusiastic, Ebullient, Lively	Full of energy and excitement

Word	Synonym(s)	Meaning
Fabricate	Construct, Manufacture, Invent	To make or create (or falsify)
Fickle	Changeable, Inconstant, Capricious	Changing frequently
Formidable	Daunting, Intimidating, Impressive	Inspiring fear or respect
Frank	Honest, Candid, Direct	Open and sincere
Frivolous	Trivial, Superficial, Silly	Not serious
Frugal	Economical, Thrifty, Sparing	Not wasteful

Furtive	Secretive, Sly, Stealthy	Attempting to avoid notice
Futile	Useless, Ineffectual, Pointless	Incapable of producing result

Word	Synonym(s)	Meaning
Gregarious	Sociable, Outgoing, Convivial	Fond of company of others
Grateful	Thankful, Appreciative, Obligated	Feeling thanks

Word	Synonym(s)	Meaning
Haphazard	Random, Disorganized, Unsystematic	Lacking any obvious organization
Haughty	Arrogant, Proud, Disdainful	Condescendingly proud

Word	Synonym(s)	Meaning
Ignorant	Uninformed, Unaware, Uneducated	Lacking knowledge
Immaculate	Spotless, Flawless, Perfect	Free from stain
Impartial	Unbiased, Fair, Objective	Not favoring one over another
Incessant	Continuous, Ceaseless, Constant	Never stopping
Innocuous	Harmless, Inoffensive, Benign	Not harmful

Word	Synonym(s)	Meaning
Jubilant	Joyful, Elated, Exultant	Feeling great joy
Judicious	Wise, Prudent, Discerning	Showing good judgment

Word	Synonym(s)	Meaning
Kindred	Similar, Related, Akin	Similar in kind or related

Word	Synonym(s)	Meaning
Lethargic	Sluggish, Lazy, Torpid	Lacking energy
Lucid	Clear, Intelligible, Coherent	Easy to understand

Word	Synonym(s)	Meaning
Malicious	Spiteful, Vindictive, Malevolent	Intending to do harm
Meticulous	Precise, Careful, Scrupulous	Very careful with details

Word	Synonym(s)	Meaning
Nonchalant	Indifferent, Casual, Unconcerned	Casually calm and relaxed

Word	Synonym(s)	Meaning
Obstinate	Stubborn, Headstrong, Persistent	Stubbornly adhering to an opinion

Ostentatious	Showy, Flashy, Pretentious	Designed to impress
--------------	----------------------------	---------------------

Word	Synonym(s)	Meaning
Placate	Appease, Pacify, Mollify	To make calm after anger
Plausible	Credible, Believable, Reasonable	Seeming reasonable
Prolific	Productive, Fertile, Abundant	Producing much or many
Prudent	Cautious, Wise, Discerning	Acting with care

Word	Synonym(s)	Meaning
Quixotic	Idealistic, Impractical, Romantic	Exceedingly idealistic

Word	Synonym(s)	Meaning
Rapid	Fast, Quick, Swift	Happening quickly
Reluctant	Unwilling, Hesitant, Disinclined	Not willing
Renowned	Famous, Celebrated, Distinguished	Known for something good
Resilient	Flexible, Strong, Adaptable	Able to recover quickly

Word	Synonym(s)	Meaning
Scrutinize	Examine, Inspect, Study	To examine carefully
Solemn	Serious, Grave, Dignified	Formal and dignified
Sporadic	Occasional, Intermittent, Irregular	Occurring at irregular intervals
Superficial	Shallow, Cursory, Surface-level	Existing only on the surface

Word	Synonym(s)	Meaning
Tactful	Diplomatic, Considerate, Discreet	Careful not to offend
Tranquil	Peaceful, Calm, Serene	Free from disturbance
Trivial	Insignificant, Minor, Petty	Of little importance
Turbulent	Chaotic, Agitated, Violent	Characterized by disorder

Word	Synonym(s)	Meaning
Unanimous	Agreed, United, Consensus	Showing complete agreement
Urbane	Sophisticated, Refined, Polished	Courteous and refined
Urgent	Pressing, Critical, Important	Requiring immediate action

Word	Synonym(s)	Meaning
Venerate	Respect, Revere, Honor	To regard with great respect
Vigilant	Watchful, Alert, Careful	Keeping careful watch

Volatile	Changeable, Unstable, Explosive	Liable to change rapidly
----------	---------------------------------	--------------------------

Word	Synonym(s)	Meaning
Wary	Careful, Cautious, Vigilant	Feeling or showing caution
Witty	Humorous, Clever, Amusing	Showing quick humor

Word	Synonym(s)	Meaning
Zealous	Enthusiastic, Ardent, Fervent	Having great energy for a cause

Part B — Antonyms Reference

Word	Antonym(s)	Word	Antonym(s)
Abate	Intensify, Escalate	Fertile	Sterile, Barren
Abhor	Love, Admire	Fickle	Steady, Constant
Abundant	Scarce, Insufficient	Frank	Dishonest, Evasive
Acute	Dull, Mild, Blunt	Frugal	Extravagant, Wasteful
Adept	Incompetent, Inept	Futile	Useful, Effective
Affluent	Poor, Impoverished	Generous	Stingy, Miserly
Ambiguous	Clear, Explicit	Genuine	Fake, Counterfeit
Ample	Insufficient, Scant	Gloomy	Cheerful, Bright
Arrogant	Humble, Modest	Grateful	Ungrateful, Thankless
Audacious	Timid, Cautious	Hardship	Comfort, Ease
Benevolent	Malevolent, Cruel	Haughty	Humble, Modest
Benign	Malignant, Harmful	Honest	Dishonest, Deceitful
Bizarre	Ordinary, Normal	Hostile	Friendly, Hospitable
Callous	Compassionate, Tender	Humble	Proud, Arrogant
Candid	Deceitful, Evasive	Ignorant	Educated, Informed
Chronic	Acute, Temporary	Immense	Tiny, Minute
Coherent	Incoherent, Disjointed	Impartial	Biased, Partial
Compassion	Indifference, Cruelty	Impossible	Possible, Feasible
Competent	Incompetent, Inept	Inadequate	Adequate, Sufficient
Concise	Long-winded, Verbose	Incompetent	Capable, Skilled
Condemn	Praise, Approve	Independent	Dependent, Reliant
Confident	Doubtful, Insecure	Innocent	Guilty, Culpable
Conspicuous	Invisible, Hidden	Intelligent	Stupid, Dull

Corrupt	Honest, Upright	Irrational	Rational, Logical
Courage	Cowardice, Fearfulness	Joy	Sorrow, Grief
Cruel	Kind, Compassionate	Judicious	Foolish, Rash
Cunning	Honest, Naive	Just	Unjust, Biased
Danger	Safety, Security	Kind	Cruel, Unkind
Deceive	Undeceive, Enlighten	Lazy	Industrious, Diligent
Defeat	Victory, Triumph	Legal	Illegal, Unlawful
Deficiency	Abundance, Surplus	Liberate	Imprison, Confine
Delicate	Sturdy, Robust	Lucid	Unclear, Confusing
Deny	Admit, Acknowledge	Magnificent	Inferior, Mediocre
Diligent	Lazy, Indolent	Mature	Immature, Childish
Diminish	Increase, Augment	Mediocre	Excellent, Superb
Disperse	Gather, Assemble	Modest	Arrogant, Conceited
Elaborate	Concise, Brief	Naive	Sophisticated, Cynical
Eloquent	Inarticulate, Halting	Narrow	Wide, Broad
Encourage	Discourage, Deter	Normal	Abnormal, Unusual
Enormous	Tiny, Minute	Obscure	Clear, Distinct
Eradicate	Establish, Implant	Optimism	Pessimism, Cynicism
Evident	Obscure, Unclear	Ordinary	Extraordinary, Unique
Exaggerate	Understate, Minimize	Plentiful	Scarce, Meager
Expand	Contract, Shrink	Powerful	Weak, Powerless
Expensive	Cheap, Inexpensive	Precious	Worthless, Cheap
Extinct	Extant, Living	Prosperity	Poverty, Hardship

Part C — Previous Year Questions

Q1 Synonym ABATE

(a) Increase (b) **Subside** (c) Aggravate (d) Escalate

Answer: (b) Subside

Q2 Synonym BENEVOLENT

(a) Cruel (b) Malevolent (c) **Kind** (d) Malicious

Answer: (c) Kind

Q3 Synonym CANDID

(a) Evasive (b) **Frank** (c) Secretive (d) Dishonest

Answer: (b) Frank

Q4 Synonym DESPISE

(a) Admire (b) Love (c) **Detest** (d) Respect

Answer: (c) Detest

Q5 Synonym ELABORATE

(a) Simple (b) **Detailed** (c) Concise (d) Brief

Answer: (b) Detailed

Q6 Synonym FURTIVE

(a) Open (b) **Secret** (c) Honest (d) Frank

Answer: (b) Secret

Q7 Synonym GREGARIOUS

(a) Antisocial (b) **Sociable** (c) Introverted (d) Reserved

Answer: (b) Sociable

Q8 Synonym HAPHAZARD

(a) Organized (b) **Random** (c) Systematic (d) Planned

Answer: (b) Random

Q9 Synonym INCESSANT

(a) Intermittent (b) **Continuous** (c) Occasional (d) Sporadic

Answer: (b) Continuous

Q10 Synonym JUBILANT

(a) Sad (b) Depressed (c) **Joyful** (d) Gloomy

Answer: (c) Joyful

Q11 Synonym LETHARGIC

(a) Energetic (b) **Sluggish** (c) Active (d) Dynamic

Answer: (b) Sluggish

Q12 Synonym MALICIOUS

(a) Kind (b) Benevolent (c) **Spiteful** (d) Gentle

Answer: (c) Spiteful

Q13 Synonym NONCHALANT

(a) Anxious (b) **Indifferent** (c) Concerned (d) Enthusiastic

Answer: (b) Indifferent

Q14 Synonym OBSTINATE

(a) Flexible (b) **Stubborn** (c) Yielding (d) Docile

Answer: (b) Stubborn

Q15 Synonym PROLIFIC

(a) Unproductive (b) **Productive** (c) Barren (d) Sterile

Answer: (b) Productive

Q16 Synonym QUIXOTIC

(a) Practical (b) **Idealistic** (c) Realistic (d) Pragmatic

Answer: (b) Idealistic

Q17 Synonym RELUCTANT

(a) Willing (b) **Unwilling** (c) Eager (d) Enthusiastic

Answer: (b) Unwilling

Q18 Synonym SUPERFICIAL

(a) Deep (b) **Shallow** (c) Profound (d) Thorough

Answer: (b) Shallow

Q19 Synonym TURBULENT

(a) Calm (b) Peaceful (c) **Chaotic** (d) Serene

Answer: (c) Chaotic

Q20 Synonym URBANE

(a) Rural (b) **Sophisticated** (c) Unrefined (d) Rustic

Answer: (b) Sophisticated

Q21 Synonym VENERATE

(a) Despise (b) **Respect** (c) Disrespect (d) Hate

Answer: (b) Respect

Q22 Synonym WARY

(a) **Careful** (b) Careless (c) Reckless (d) Foolish

Answer: (a) Careful

Q23 Synonym ZEALOUS

(a) Indifferent (b) Apathetic (c) **Enthusiastic** (d) Lethargic

Answer: (c) Enthusiastic

Q24 Synonym ACUMEN

(a) Stupidity (b) Foolishness (c) **Insight** (d) Ignorance

Answer: (c) Insight

Q25 Antonym ABATE

(a) Diminish (b) Subside (c) **Intensify** (d) Decline

Answer: (c) Intensify

Q26 Antonym BENEVOLENT

(a) Kind (b) Charitable (c) **Malevolent** (d) Generous

Answer: (c) Malevolent

Q27 Antonym CANDID

(a) Frank (b) Honest (c) **Evasive** (d) Direct

Answer: (c) Evasive

Q28 Antonym DESPISE

(a) Hate (b) Detest (c) **Admire** (d) Loathe

Answer: (c) Admire

Q29 Antonym ELABORATE

(a) Detailed (b) Intricate (c) **Concise** (d) Complex

Answer: (c) Concise

Q30 Antonym FRUGAL

(a) Economical (b) Thrifty (c) **Extravagant** (d) Sparing

Answer: (c) Extravagant

Q31 Antonym GLOOMY

(a) Dark (b) **Cheerful** (c) Dreary (d) Melancholy

Answer: (b) Cheerful

Q32 Antonym HONOR

(a) Respect (b) Esteem (c) **Disgrace** (d) Glory

Answer: (c) Disgrace

Q33 Antonym IMMACULATE

(a) Spotless (b) **Flawed** (c) Pristine (d) Perfect

Answer: (b) Flawed

Q34 Antonym JUDICIOUS

(a) Wise (b) Prudent (c) **Rash** (d) Discerning

Answer: (c) Rash

Q35 Antonym LAVISH

(a) Generous (b) **Frugal** (c) Extravagant (d) Munificent

Answer: (b) Frugal

Q36 Antonym MODEST

(a) Humble (b) Unassuming (c) **Arrogant** (d) Meek

Answer: (c) Arrogant

Q37 Antonym NAIVE

(a) Innocent (b) Artless (c) **Sophisticated** (d) Simple

Answer: (c) Sophisticated

Q38 Antonym OBSCURE

(a) Vague (b) Unclear (c) **Clear** (d) Ambiguous

Answer: (c) Clear

Q39 Antonym PROFOUND

(a) Deep (b) **Shallow** (c) Intense (d) Thorough

Answer: (b) Shallow

Q40 Antonym RELUCTANT

(a) Unwilling (b) Hesitant (c) **Eager** (d) Disinclined

Answer: (c) Eager

Q41 Antonym STERN

(a) Strict (b) Severe (c) **Lenient** (d) Harsh

Answer: (c) Lenient

Q42 Antonym TRANQUIL

(a) Peaceful (b) Calm (c) **Turbulent** (d) Serene

Answer: (c) Turbulent

Q43 Antonym URBANE

(a) Sophisticated (b) Refined (c) **Rustic** (d) Polished

Answer: (c) Rustic

Q44 Antonym VIGILANT

(a) Watchful (b) Alert (c) **Careless** (d) Cautious

Answer: (c) Careless

Q45 Antonym WITHDRAW

(a) Remove (b) Extract (c) **Advance** (d) Retire

Answer: (c) Advance

Q46 Antonym ZEAL

(a) Passion (b) **Apathy** (c) Enthusiasm (d) Fervor

Answer: (b) Apathy

Q47 Antonym ABUNDANT

(a) Plentiful (b) Copious (c) **Scarce** (d) Ample

Answer: (c) Scarce

Q48 Antonym COMPASSION

(a) Sympathy (b) Empathy (c) **Indifference** (d) Kindness

Answer: (c) Indifference

Quick Practice

#	Question	Answer
Q1	Synonym of AUDACIOUS	(b) Bold
Q2	Antonym of BRIEF	(c) Lengthy
Q3	Synonym of CURIOUS	(b) Inquisitive
Q4	Antonym of DILIGENT	(c) Lazy
Q5	Synonym of EMANATE	(a) Originate
Q6	Antonym of FRANK	(c) Evasive
Q7	Synonym of GRATIFY	(b) Satisfy

Q8	Antonym of HUMBLE	(b) Arrogant
Q9	Synonym of INNOVATIVE	(c) Groundbreaking
Q10	Antonym of JOYFUL	(c) Sorrowful

Part D — Exam Strategy for Vocabulary

1. **Daily reading habit** — read editorials in The Hindu / Indian Express; note new words in context.
2. **Flashcards** — build word cards (Anki, Quizlet, or physical) and revise with spaced repetition.
3. **Learn roots, prefixes & suffixes** — bene- (good), male- (bad), anti- (against), multi- (many), -logy (study of), -phobia (fear of).
4. **Pair synonyms with antonyms** when learning a new word (e.g. Abate → Diminish → Intensify).
5. **Practice PYQs** — SSC CGL repeats high-frequency words like Abate, Benevolent, Candid.
6. **Elimination method** — rule out obviously wrong options first when unsure of exact meaning.
7. Watch for **commonly confused word pairs**: Accept/Except, Affect/Effect, Compliment/Complement, Desert/Dessert, Principal/Principle, Their/There/They're.

Static GK: Medieval India

Delhi Sultanate & Mughal Empire — rulers, reforms, battles, architecture, and PYQs.

Medieval Indian History is a highly scoring topic in the SSC CGL General Awareness section (typically **3-5 questions**). Focus areas: Delhi Sultanate (1206-1526), Mughal Empire (1526-1857), key battles, art & architecture, and administration.

Part A — Delhi Sultanate (1206-1526 AD)

Dynasty	Period	Founder	Key Rulers
Slave (Mamluk)	1206-1290	Qutb-ud-din Aibak	Iltutmish, Razia, Balban
Khilji	1290-1320	Jalal-ud-din Khilji	Ala-ud-din Khilji
Tughlaq	1320-1414	Ghiyas-ud-din Tughlaq	Muhammad bin Tughlaq, Firuz Shah Tughlaq
Sayyid	1414-1451	Khizr Khan	—
Lodi	1451-1526	Bahlol Lodi	Sikandar Lodi, Ibrahim Lodi

Slave (Mamluk) Dynasty

Ruler	Reign	Key Facts
Qutb-ud-din Aibak	1206-1210	Titled "Lakh Baksh"; started Qutub Minar; built Quwwat-ul-Islam Mosque (first mosque in India); died playing polo (Chaugan)
Iltutmish	1211-1236	Real founder of the Sultanate; introduced Silver Tanka & Copper Jital; organized Chahalgani (Group of Forty); completed Qutub Minar; father of Razia Sultana
Razia Sultana	1236-1240	First & only Muslim woman ruler of Delhi; insisted on the title "Sultan"; defeated and killed by her brother Bahram Shah
Balban	1266-1287	Introduced Theory of Kingship; introduced Sijda & Paibos (Persian court customs); broke power of Chahalgani; strong spy network

Khilji Dynasty

★ Ala-ud-din Khilji (1296-1316) — most important ruler

Introduced price-control market reforms via Diwan-i-Riyasat and Shahna-i-Mandi (market inspectors); fixed land revenue at 50% of produce (Mauza system); repelled multiple Mongol invasions; introduced Dagh (horse branding) and Chehra (soldier descriptive rolls); general Malik Kafur conquered Devagiri, Warangal, and Madurai in the Deccan.

Term	Meaning
Diwan-i-Riyasat	Department of Commerce
Shahna-i-Mandi	Market Controller

Dagh	Branding of horses
Chehra	Descriptive roll of soldiers

Tughlaq Dynasty

★ Muhammad bin Tughlaq (1325-1351) — "The Mad Sultan"

Four famous failed experiments:

1. **Capital transfer:** Delhi → Daulatabad (Devagiri), later reversed — massive hardship and desertions.
2. **Token currency:** copper/brass coins given silver-tanka legal value — led to counterfeiting, withdrawn.
3. **Increased land tax** in the Doab region during a famine — caused rebellion.
4. **Qarachil expedition** to the Himalayan region — army destroyed.

Ibn Battuta, the Moroccan traveler, visited India during his reign and documented these events.

Ruler	Key Contribution
Ghiyas-ud-din Tughlaq	Founder of the dynasty; died when a pavilion collapsed
Muhammad bin Tughlaq	Capital transfer, token currency, Qarachil expedition
Firuz Shah Tughlaq	Built canals (Sutlej, Yamuna), hospitals, colleges, Feroz Shah Kotla; first to impose Jizya even on Brahmins

Sayyid & Lodi Dynasties

Ruler	Key Facts
Sayyid Dynasty (Khizr Khan)	Weakest & shortest dynasty; ruled as vassals of the Timurids
Bahlol Lodi	Founder of Lodi dynasty; conquered Gwalior, Jaunpur, Bihar
Sikandar Lodi	Shifted capital Delhi → Agra; imposed Jizya; built Moth ki Masjid
Ibrahim Lodi	Last Sultan of Delhi; nobles rebelled; Daulat Khan Lodi invited Babur; defeated at the First Battle of Panipat (1526) — end of the Sultanate

Part B — Mughal Empire (1526-1857 AD)

Ruler	Reign	Significance
Babur	1526-1530	Founder; First Battle of Panipat
Humayun	1530-1540, 1555-1556	Lost and regained the empire
Akbar	1556-1605	Greatest Mughal Emperor
Jahangir	1605-1627	Fine arts, East India Company trading rights
Shah Jahan	1628-1658	"Builder King" (Taj Mahal)

Aurangzeb	1658–1707	Last powerful Mughal; empire's decline began
-----------	-----------	--

★ Babur (1526–1530) — Founder

Battle	Year	Opponent	Result
First Battle of Panipat	1526	Ibrahim Lodi	Babur won — founded the Mughal Empire; first use of artillery & gunpowder in India
Battle of Khanwa	1527	Rana Sanga of Mewar	Babur won — ended Rajput resistance
Battle of Chanderi	1528	Medini Rai	Babur won
Battle of Ghagra	1529	Afghan Chiefs	Babur won

Babur wrote his memoirs *Tuzuk-i-Baburi* (Baburnama) in Turkish, describing India's flora, fauna, and people.

Humayun (1530–1540, 1555–1556)

Battle	Year	Opponent	Result
Battle of Chausa	1539	Sher Shah Suri	Humayun defeated
Battle of Kannauj	1540	Sher Shah Suri	Humayun defeated; fled to Persia

Regained the throne in 1555 (defeated Sikandar Suri at Sirhind) with Persian help from Shah Tahmasp. Built Dinpanah in Delhi. Died in 1556, falling from the stairs of his library.

★ Akbar (1556–1605) — Greatest Mughal Emperor

Became emperor at age 13 (regent: Bairam Khan, 1556–1560). Won the **Second Battle of Panipat (1556)** against Hemu.

Battle	Year	Opponent	Result
Battle of Malwa	1561	Baz Bahadur	Akbar won
Battle of Chittor	1567–68	Maharana Udai Singh	Akbar won
Battle of Ranthambore	1568	—	Captured by Akbar
Battle of Gujarat	1573	Mirza Hakim	Akbar won
Battle of Haldighati	1576	Maharana Pratap	Akbar won (led by Man Singh)

Religious Policy

Abolished **Jizya**; established **Ibadat Khana** (House of Worship) at Fatehpur Sikri; formulated **Din-i-Ilahi** (Divine Faith, blending Islam, Hinduism, Christianity — only ~18 converts, including Birbal); known for **Sulh-i-Kul** (religious tolerance).

Administration & Architecture

Mansabdari system — military rank system (ranks 10–5000), paid in cash (Naqd) or land (Jagir). Built **Fatehpur Sikri** (with Buland Darwaza, Jama Masjid, Panch Mahal, Diwan-i-Khas, Diwan-i-Aam) and the **Agra Fort**.

Navratnas (Nine Gems)

Name	Role
Birbal	Hindu advisor, known for wit
Tansen	Greatest musician of the era
Abul Fazal	Wrote Akbarnama & Ain-i-Akbari
Faizi	Poet and intellectual
Raja Man Singh	Rajput general
Raja Todar Mal	Revenue minister; introduced Bandobast system
Abdul Rahim Khan-e-Khana	Poet
Mulla Do Pyaza	Advisor
Fakir Aziao-Din	Spiritual advisor

Term	Meaning
Sulh-i-Kul	Absolute peace / religious tolerance
Ibadat Khana	House of Worship
Din-i-Ilahi	Divine Faith
Mansabdari	Military rank system
Bandobast	Land revenue system
Zabti System	Land measurement & revenue assessment

Jahangir (1605–1627)

Titled Nur-ud-din Muhammad Jahangir. Great patron of Mughal miniature painting; introduced Pietra Dura (colored stone inlay, later used in the Taj Mahal). Granted trading rights to the **British East India Company in 1615** (Sir Thomas Roe led the English mission). Established the **Chain of Justice**. Wife Nur Jahan was a powerful, de facto ruler.

★ Shah Jahan (1628–1658) — "Builder King"

Structure	Details
Taj Mahal	Built in Agra in memory of wife Mumtaz Mahal; architect Ustad Ahmad Lahori; white marble with Pietra Dura
Red Fort (Lal Qila)	Built in Delhi — replaced Agra as capital
Jama Masjid	Built in Delhi

Peacock Throne (Takht-e-Taus)	Built in 1635, cost ~10 crore rupees
Moti Masjid	Built in Agra Fort (Pearl Mosque)
Shalimar Gardens	Built in Lahore
Shish Mahal	Built in Lahore Fort (Mirror Palace)

War of succession: Aurangzeb (3rd son) defeated and killed Dara Shikoh (eldest son), then deposed Shah Jahan in 1658. Shah Jahan was imprisoned in Agra Fort until his death in 1666, viewing the Taj Mahal from his window.

Aurangzeb (1658-1707) — Last Powerful Mughal

Titled **Alamgir** ("World Conqueror"). Re-imposed **Jizya**; destroyed temples at Kashi, Mathura, Somnath; abolished Sulh-i-Kul. Expanded the empire to its greatest territorial extent (Deccan campaigns) but fought a prolonged, costly war against the Marathas (Shivaji, Sambhaji). Faced the Jat rebellion and Sikh revolts (under Guru Tegh Bahadur and Guru Gobind Singh). His orthodox policies are widely seen as the trigger for the empire's decline after his death in 1707.

Part C — Important Reference Tables

Sources of Medieval Indian History

Source	Author	About
Tarikh-i-Firoz Shahi	Zia-ud-din Barani	Tughlaq rulers
Futuh-us-Salatin	Isami	Delhi Sultanate
Kitab-ul-Rihla	Ibn Battuta	Travelogue of India
Tuzuk-i-Baburi	Babur	Autobiography (Turkish)
Akbarnama	Abul Fazal	Official history of Akbar's reign
Ain-i-Akbari	Abul Fazal	Akbar's administration
Padshahnama	Abdul Hamid Lahori	Shah Jahan's reign

Important Taxes

Tax	Meaning
Kharaj	Land revenue (1/5 to 1/2 of produce)
Jizya	Tax on non-Muslims (abolished by Akbar, re-imposed by Aurangzeb)
Zakat	Religious tax on Muslims (2.5% of savings)
Khams	War booty (1/5 share for the state)
Sharb	Irrigation tax
Charai	Grazing tax on cattle

Important Battles — Quick Reference

Battle	Year	Fought Between	Result
First Battle of Panipat	1526	Babur vs Ibrahim Lodi	Babur won → Mughal Empire began
Battle of Khanwa	1527	Babur vs Rana Sanga	Babur won
Battle of Chausa	1539	Humayun vs Sher Shah Suri	Sher Shah won
Battle of Kannauj	1540	Humayun vs Sher Shah Suri	Sher Shah won
Second Battle of Panipat	1556	Akbar vs Hemu	Akbar won
Battle of Haldighati	1576	Akbar vs Maharana Pratap	Akbar won
Battle of Plassey	1757	Siraj-ud-daulah vs British	British won — beginning of British rule
Battle of Buxar	1764	Mughal vs British	British won

Part D — Previous Year Questions

Q1 First ruler of the Delhi Sultanate?

(a) Iltutmish (b) **Qutb-ud-din Aibak** (c) Balban (d) Ala-ud-din Khilji

Answer: (b)

Q2 Tanka & Jital coins introduced by?

(a) Aibak (b) Balban (c) **Iltutmish** (d) Ala-ud-din Khilji

Answer: (c)

Q3 Only Muslim woman ruler of Delhi?

(a) Nur Jahan (b) Mumtaz Mahal (c) **Razia Sultana** (d) Jodha Bai

Answer: (c)

Q4 Ala-ud-din Khilji introduced which reform?

(a) **Price control system** (b) Token currency (c) Land revenue system (d) Mansabdari

Answer: (a)

Q5 Who introduced Token Currency?

(a) Ala-ud-din Khilji (b) **Muhammad bin Tughlaq** (c) Firuz Shah Tughlaq (d) Iltutmish

Answer: (b)

Q6 First Battle of Panipat (1526) fought against?

(a) Rana Sanga (b) **Ibrahim Lodi** (c) Hemu (d) Sher Shah Suri

Answer: (b)

Q7 Who wrote 'Tuzuk-i-Baburi'?

(a) Humayun (b) Abul Fazal (c) **Babur** (d) Akbar

Answer: (c)

Q8 Second Battle of Panipat won by?

(a) Babur (b) Humayun (c) **Akbar** (d) Shah Jahan

Answer: (c)

Q9 Akbar abolished which tax?

(a) Kharaj (b) Zakat (c) **Jizya** (d) Sharb

Answer: (c)

Q10 Established the Ibadat Khana?

(a) Babur (b) **Akbar** (c) Jahangir (d) Aurangzeb

Answer: (b)

Q11 NOT a Navratna of Akbar?

(a) Birbal (b) Tansen (c) Abul Fazal (d) **Nur Jahan**

Answer: (d)

Q12 Taj Mahal built by?

(a) Akbar (b) Jahangir (c) **Shah Jahan** (d) Aurangzeb

Answer: (c)

Q13 Granted rights to British East India Co.?

(a) Akbar (b) **Jahangir** (c) Shah Jahan (d) Aurangzeb

Answer: (b)

Q14 Re-imposed Jizya on non-Muslims?

(a) Akbar (b) Jahangir (c) Shah Jahan (d) **Aurangzeb**

Answer: (d)

Q15 Peacock Throne built by?

(a) Akbar (b) Jahangir (c) **Shah Jahan** (d) Aurangzeb

Answer: (c)

Q16 Known as 'Lakh Baksh'?

(a) Iltutmish (b) **Qutb-ud-din Aibak** (c) Balban (d) Ala-ud-din Khilji

Answer: (b)

Q17 Also called the Mamluk dynasty?

(a) Khilji (b) Tughlaq (c) **Slave** (d) Lodi

Answer: (c)

Q18 Last ruler of the Delhi Sultanate?

(a) Bahlol Lodi (b) Sikandar Lodi (c) **Ibrahim Lodi** (d) Khizr Khan

Answer: (c)

Q19 Who wrote 'Akbarnama'?

(a) **Abul Fazal** (b) Faizi (c) Birbal (d) Abdul Rahim Khan-e-Khana

Answer: (a)

Q20 Chain of Justice established by?

(a) Akbar (b) **Jahangir** (c) Shah Jahan (d) Aurangzeb

Answer: (b)

Q21 Maharana Pratap defeated at?

(a) Chittor (b) **Haldighati** (c) Ranthambore (d) Gujarat

Answer: (b)

Q22 Regent of Akbar in early years?

(a) Abul Fazal (b) Birbal (c) **Bairam Khan** (d) Raja Man Singh

Answer: (c)

Q23 Bandobast system introduced by?

(a) Birbal (b) Tansen (c) **Raja Todar Mal** (d) Abul Fazal

Answer: (c)

Q24 Father of Razia Sultana?

(a) Aibak (b) **Iltutmish** (c) Balban (d) Ala-ud-din Khilji

Answer: (b)

Q25 Battle of Plassey marked the start of?

(a) Mughal Empire (b) Maratha Empire (c)

British Empire (d) French Empire

Answer: (c)

Quick Revision — Practice Questions

#	Question	Answer
Q1	Who started the construction of Qutub Minar?	Qutb-ud-din Aibak
Q2	Who introduced the Mansabdari system?	Akbar
Q3	Book written by Abul Fazal about Akbar's administration?	Ain-i-Akbari
Q4	Which Mughal emperor was the 'Builder King'?	Shah Jahan
Q5	Who was the regent of Akbar?	Bairam Khan
Q6	Who built the Red Fort in Delhi?	Shah Jahan
Q7	Which tax was re-imposed by Aurangzeb?	Jizya
Q8	Who was the first woman ruler of Delhi?	Razia Sultana
Q9	Who introduced the Token Currency?	Muhammad bin Tughlaq
Q10	Which battle ended the Delhi Sultanate?	First Battle of Panipat (1526)